

Overview and Objectives

The objective of this course is to provide an introduction to computer organization, processors, instruction sets, and assembly language programming.

This course will cover programming in the MIPS assembly language, with a number of programming assignments using the QtSPIM simulator. The basic architecture of a processor, data representation, and micro-code will be the main topics of this class along with the assembly language programming.

At the end of this course the student should be able to

- Understand the different components of a computer and explain how they interact.
- Read and write MIPS assembly code using the integer registers, including implementing and calling functions.
- Understand the binary representations of integers and floating point numbers in a computer, translating base ten representations to and from binary representations.
- Understand the representation of computer instructions in a computer, translating from machine code to and from assembly code.
- Understand the binary representations of text, music, images, and movies in a computer.
- Understand how math operations are carried out in binary and how this is implemented in the computer's electronic circuitry.
- Understand how computer components are constructed from electronic circuits.
- Follow a computer instruction through a CPU's datapath and explain how the circuits are executing the instruction.
- Explain how datapath pipelining works and the complications that arise.
- Understand the reason for and design of a memory hierarchy including caching and virtual memory.
- Explain the differences between various kinds of parallel processing.

Essential Information

Instructor: Erik Steinmetz
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Office: Hagfors 369E
Office Hours: MW 1:30 pm to 2:00 pm, Th 9:00 am to 9:30 am, or by appointment
Class Location: Hagfors 352
Class Times: MWF 9:40 am to 10:40 am
Class Dates: 3 September through 12 December, 2025
OER Text: "MIPS Assembly Language Programming using QtSpim"; Jorgensen
"Computer Organization with MIPS"; Bergmann
Prerequisites: Software Design and Development (CSC 321), Data Structures (CSC 341)
and Discrete Mathematics (MAT 202 or MAT 302) with a C- or better.

Grading

Grades will be determined based on the accumulation of up to 1000 points. There will be a number of quizzes and a final quiz adding up to a total of 500 points. 300 points will be given for the completion of various coding projects. The remaining 200 points are awarded for other homework and participation in class.

The projects will be written up using the MIPS assembly language. As always, your code should:

- Be properly commented.
- Work correctly.
- Be tested over different inputs.

Homework must be submitted in a legible manner. Plain text or pdf formats are required.

Late work will not be accepted. Turn your work in before the deadline even if it is not complete. On some assignments you will be allowed a single redo to improve your comprehension of the material and your score, but only if you have submitted a reasonable first attempt on time.

The entire one thousand points are tallied at the end of the semester. Typically $total \geq 900$ will be various flavors of A grades, $total \geq 800$ B grades, and $total \geq 700$ C grades. $700 > total > 550$ will result in a D grade, while lower will be an F. Regardless of overall accumulated points, receiving less than 70% of the quiz points will automatically result in a D grade or lower.

Please note that a C- or higher is required for a pass under the pass / no-pass grading option. A passing grade, C- or higher, is required for the course to apply to the Computer Science major or minor or the Mathematics major.

Expectations

- **Attend every class session.** Most of the class sessions will be lecture and activities, while some class sessions will be given over to lab work (writing code).
- **Use class time effectively and allow others to do the same.** Our class time is for discussing, reading, learning, and writing about what goes on inside the computer. It is not time for surfing social networking sites, watching cat videos, or texting friends. No use of phones is allowed in class other than taking photos of the whiteboard. No use of laptops other than writing code or for approved accommodations is permitted.
- **Submit assignments and code when due.** Much of your grade in this course is dependent on the timely completion of assignments and coding projects. Turn these in on time for the best results.
- **Write your own code.** All the work you submit, whether code or other assignments, must be your own work. You are encouraged to discuss ideas, solutions, explanations, and references with your classmates. You are also encouraged to work together as you do assignments and write code, but everything you submit must be your own handiwork. Please refer to the Augsburg Academic Honesty Policy which can be found at <https://tinyurl.com/AcademicHonestyAugsburg>. Please note that penalties for breaching academic honesty policies include receiving a zero on the assignment and may result in receiving an F in the course. To establish a clear division between acceptable and unacceptable sharing of code, no electronic copying of code is allowed: you must type everything in yourself. Do not send or receive code from your classmates and do not copy-paste code from any other source. Both sender and receiver of copied materials are considered in violation of the no-copying rule.
- **Schedule your time.** Reading materials from the text, working on assignments, and writing the code for this course is time consuming. Expect to spend six hours or more each week reading, writing, and coding outside of classtime to succeed in this class.
- **Pay attention to Moodle and email.** Our schedule and assignments are posted on Moodle and may be updated there. I will send email notifying you of any significant changes or updates.

Schedule of Topics

We will cover the following topics along with coverage of reading and writing code in MIPS assembly language.

- Overview and history of computing hardware.
- Basic MIPS assembly language instructions.
- Binary representations of integer values.
- Binary representations of floating point values.
- MIPS Machine code - binary representation of instructions.
- Advanced MIPS assembly language instructions.
- Binary representations of text.
- Binary representations of music, images, and movies.
- Math and hardware implementations.
- Building logical circuits from transistors.
- Building computer components from logical circuits.
- Creating a CPU datapath from computer components.
- Creating a pipelined datapath.
- Cache and Virtual Memory.
- Parallel Processing Implementations.

Additional Information

Disability Accommodations and Accessibility: Students with disabilities who believe that they may need accommodations in this class are encouraged to contact the Center for Learning and Adaptive Student Services (CLASS) (612-330-1053 or stop by the Gage Center welcome desk) as soon as possible to better ensure that such accommodations are implemented in a timely fashion. Contact info about the CLASS program can be found at <http://www.augsburg.edu/class/>. Be aware that it is your responsibility to meet the requirements of your CLASS office accommodations.

Disclaimer: Policies stated within this syllabus are subject to change, following verbal announcement in class. Students are responsible for knowing the current version, always to be found on the course Moodle page.